

BASF

We create chemistry

Laricyl®

Pore minimizer

Global Except US

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SUMMARY FILE

Laricyl	BC10206
Description	Extract of organic *Fomes officinalis* mushroom mycelium, grown by solid state fermentation
Regulatory data	
INCI	Butylene Glycol (and) Fomes Officinalis (Mushroom) Extract (and) PEG-40 Hydrogenated Castor Oil
China	All components are listed in Inventory of Existing Cosmetic Ingredient in China (IECIC 2021)
CAS#	107-88-0; 94465-74-4; 61788-85-0
Appearance	Colorless to yellow, clear syrupy liquid with weak characteristic odor
Preservative	None
Natural label	None
Naturalness content (ISO 16128)	99.7% from Natural Origin
Cosmetic use	
Properties	Helps to minimize the pore appearance, while contributing to reduce shine and to maintain skin moisture
Applications	Face care, astringent, toner
Formulation data	
Concentration of use	3%
Solubility	Soluble in water, insoluble in oils and fats
Incorporation method	Incorporate Laricyl into the cosmetic product below 50°C, during the final process or at room temperature for cold processing.
Optimal pH	6-8



Unhappy with your large pores?

People commonly complain about their large visible skin pores, mostly located on the face (nose, cheeks, etc), which affect their beauty.

They find them too dilated and mostly associated with oily and over-shinning skin. For example in Europe 31% of consumers have such a concern [Men's and Women's Facial Skincare - Europe – 2018]. Enlarged facial pores are known to be a significant cosmetic concern in Asian women. One large Japanese survey estimated that more than half of women ages 20–30 complained of conspicuous pores [Katsuta *et al*, 2004].

Enlarged facial pores are the visible, topographic feature that corresponds to enlarged openings of pilosebaceous follicles. They appear either as empty funnels or filled with cornified plugs, *i.e.* comedones. Many endogenous and exogenous factors have been reported to cause enlarged pilosebaceous pores. Such factors include sex, genetic predisposition, hygiene, seasons, relative humidity and temperature, ageing, chronic ultraviolet light exposure, comedogenic xenobiotics, acne and seborrhoea. Enlarged pore sizes are associated with increased sebum output level and age [Roh *et al*, 2006]. In female, additional hormonal factors, such as those of the menstrual cycle, contribute to worsen the pore size. Pore area and volume have been shown to increase with age [Roh *et al*, 2006; Murakami *et al*, 2006]. Pores tend to open with aging and do not appear to close again in old skin [Mizukoshi and Takahashi, 2014]. In another multi-ethnic study on 2585 women, Flament and coworkers have reported that the pore size and density varied greatly with ethnicity, plateaued with age in most cases, and globally reflected self-perception by the subjects [Flament *et al*, 2015].

Laricyl: Extract of *Fomes officinalis* mushroom mycelium grown by solid state fermentation

Laricyl, with the reference LS8865, was our 20 years tried-and-tested ingredient extracted from the *Fomes officinalis* mushroom, for pore tightening. It was produced by extraction of the mushroom conks harvested from the wild in Russia on the bark of larch trees.

However, *F. officinalis* population declined with a reduction of a 70-75% population in the past caused by habitat degradation and collection for medicinal purposes. Because of its valuable biological properties and its use in 18th and 19th-century pharmacy, *F. officinalis* used to be mass-collected [Muszynska *et al*, 2020]. In March 2019, it has been assessed as an endangered species by IUCN (International Union for Conservation of Nature) and listed in the Red List as Endangered under criteria A2ad [Kalucka & Svetasheva, 2019]. This species is considered to be rare due to the combination of slow growth, habitat destruction and over harvesting of the fruiting body.

Protecting natural resources is one of three focus areas of the company's Responsibly Active program for its bioactive portfolio. Among other things, the program seeks to have 100% of botanical supply chains traceable to the source and sustainably sourced as well as 50% organic certified by 2030.

According to this commitment, BASF BCS had initiated in 2018 a project to replace the wild sourcing of Laricyl by a more sustainable cultivation by solid state fermentation.



Fungi description

Taxon name: *Fomitopsis officinalis* (Vill.) Bondartsev & Singer

Synonyms: *Fomes officinalis* (Vill.) Bres., *Laricifomes officinalis* (Vill.) Kotl. & Pouzar, *Ungulina officinalis* (Vill.) Pat., *Agaricum officinale* (Vill.) Donk, *Boletus laricis* (F. Rubel), *Boletus officinalis* Villars, *Cladomeris officinalis* (Vill.) Quél., *Leptoporus sulphureus* var. *officinalis* (Vill.) Qué, *Polyporus laricis* (F. Rubel) Duby, *Polyporus officinalis* (Vill.)

Common name: Agarikon, Eburiko, polypore du mélèze, polypore officinal

Taxonomy:

Kingdom	Phylum	Class	Order	Family
Fungi	Basidiomycota	Agaricomycetes	Polyporales	Fomitopsidaceae

Part used: mushroom mycelium

Description:
Fomes officinalis is a wood-inhabiting parasitic fungus forming distinctive conks. It is widely distributed in the northern Hemisphere, mostly in subalpine, montane hypoarctic and boreal old-growth forests on large and old coniferous trees such as larch (*Larix decidua*) but also known on Pinus, Abies and Cedrus.
Its habitats may be found in Africa (Morocco), Asia (China, India, Japan, Korea, Mongolia, Russia, Turkey), North America (USA, Canada) and Europe (Austria, Belarus, France, Greece, Spain, The Netherlands, Lithuania, Germany, Poland, Russia, Romania, Slovakia, Slovenia, Switzerland, Ukraine and Italy). Currently, the species is most commonly found in Siberia, from the foothills of the Ural Mountains to the shores of the Pacific Ocean. In Europe, *F. officinalis* resides on old larches, found mostly in natural forests [Grienke *et al*, 2014; Muszynska *et al*, 2020].

Traditional medicinal uses and phytochemistry

Mushrooms such as *F. officinalis* have a long history of uses in various folk medicines such as in Asia, Russia, the USA, Canada, Mexico, and Venezuela and are extensively applied in Traditional Chinese Medicine [Grienke *et al*, 2014]. In ancient Greece and Rome, *F. officinalis* was widely-known under the name of 'agarikon' or 'agaricum', mostly as an anti-tuberculosis agent. It is considered as one of the species of medicinal mushrooms most commonly used in traditional European medicine, used to make a bitter herbal medicine with a wide range of uses such as treatment of pain, inflammation, jaundice, fever, insect bites, muscular diseases, bladder problems, sciatica, rheumatism and hemorrhoids [Muszynska *et al*, 2020]. It was also reported for curbing sweating, astringency or its tightening properties [Grienke *et al*, 2014]. For indigenous people of the northwestern coast region of North America and Canada, this fungus is known as "Bread of Ghosts" with important spiritual as well as a medicinal role in their society. They were believed to have supernatural powers facilitating contact with spirits [Muszynska *et al*, 2020].

F. officinalis has been reported to contain several bioactive compounds with anti-inflammatory and anti-microbial properties, such as triterpenoids, polysaccharides, organic acids, coumarins and phenolic compounds [Muszynska *et al*, 2020; Girometta, 2019; Grienke *et al*, 2014].

Sourcing: biotechnological process of solid state fermentation

Solid state fermentation is defined as technique for growing microorganisms on solid, non-soluble materials in the absence or near absence of free water [Bhargav *et al*, 2008, Thomas *et al*, 2013]. Fungi are the best type of microorganisms for solid state fermentation as this biotechnological culture process mimics their natural habitat. Indeed, fungi typically grows in nature on solid substrates such as soil, piece of woods, barks… low in moisture.

The major advantages of this biotechnological culture process over harvesting from wild is to have no impact on the species population, and controlled and reproductible conditions of production (sterilized substrate, cleanroom, temperature control). Over liquid or submerged fermentation, the major advantage of solid state fermentation is its low environmental footprint with low or no water consumption and lower energy requirements [Thomas *et al*, 2013].

In solid substrate fermentation, the process occurs in absence or near absence of free water by employing a natural substrate as solid support. It brings cultivated fungi in tight contact with the insoluble substrate to achieve the highest nutrient concentration from the substrate for fermentation [Bhargav *et al*, 2008]. This mode of culture has the advantage to accelerate the growth of the fungus compared to submerged fermentation and natural environment where the growth is very slow.

Processing (Fig. 1)

The organic nutrients, millet grains (origin: Austria, Italy or Germany), gypsum, lime (origin: Austria) and some water are combined in bags to form the solid substrate for the mushroom mycelium cultivation. After sterilization by autoclaving at 121°C, the solid substrate is inoculated with the mushroom mycelium (origin: strain collected from wild in Russia in 1981) and the bags are incubated under controlled conditions of temperature, moisture, CO_2 and light, until the complete nutrient consumption and conversion into mushroom mycelium. The duration of this biotechnological production is of approximately 4 months.

After the solid state fermentation, the organic certified mushroom mycelium is dried at temperature below 40°C to protect heat sensitive ingredients, and grinded in a fine powder before extraction using a combination of water and butylene glycol as solvent and 0.3% of PEG-40 Hydrogenated Castor Oil as solubilizer. Laricyl BC10206 is obtained after this extraction step followed by separation, filtration and quality control steps.

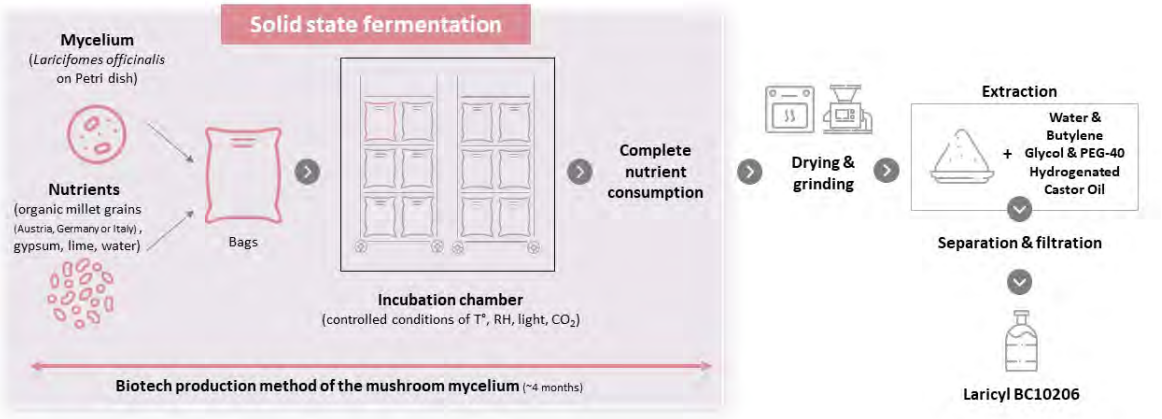


Figure 1 - Laricyl BC10206 process.

Safety/Tolerability

Laricyl was evaluated to ensure its safety in the recommended conditions of use.

Description

Laricyl is an extract of organic certified *Fomes officinalis* mushroom mycelium, grown by solid state fermentation, available as a preservative-free liquid

INCI name: Butylene Glycol (and) Fomes Officinalis (Mushroom) Extract (and) PEG-40 Hydrogenated Castor Oil

Naturalness content (according to ISO 16128): 99.7% from natural origin

Recommended dose of use: 3%

China IECIC listed

Vegan Friendly **



Illustration of a sample of Laricyl BC10206 (center), incorporated at 3% in an hydrogel (on the left) and an emulsion (on the right).

** No involvement of animal genes or animal-derived substances; unlikely cross-contamination from animal substances; no animal testing has been carried out by or on behalf of BASF on the ingredients of the product after 11th March 2009 and/or after the extended deadline 11th March 2013 for the purposes of the Cosmetic Regulation (EU) 1223/2009 (statement available upon request).



DEMONSTRATED EFFICACY

The properties of Laricyl were studied *in vivo* and *ex vivo* for:

- pore tightening (*in vivo* study),
- astringent feeling sensation (*in vivo* study),
- excess oil removal (*in vivo* study),
- moisturization (*ex vivo* study).

EFFICACY

Pore tightening: Laricyl minimizes visible pores

in vivo

OBJECTIVE

The clinical study was run to evaluate the pore tightening efficacy of the new generation of Laricyl BC10206 (from solid state fermentation process) at 3% as compared to the previous Laricyl LS8865 (from mushroom harvested from wild), immediately following application on the face.
The study was a double blind, split-face and randomized comparative study on 25 female subjects having visible pores and oily skin. The pore-tightening efficacy of Laricyl BC10206 was evaluated using VISIA Imaging and image analysis. Consumer self-perception questionnaires regarding product efficacy were also administered. Illustrative macro-images of pores were also taken on 2 additional oily-skinned volunteers with large visible pores.

RESULTS & DISCUSSION

Immediately after application, the new Laricyl BC10206 showed a significant decrease ($p<0.01$) by 5.4% in total pore area as compared to baseline, which is also significantly ($p<0.05$) 2.8 times better than the previous Laricyl generation (Fig. 2A). On the pore count parameter, similar results were obtained with a significant decrease ($p<0.01$) by 4.9%, which is also significantly 2.3 times better than the previous Laricyl generation.

These results by image analysis were confirmed by the volunteer's self-assessment (Fig. 2B). A significant majority ($p<0.05$) of 76% of the volunteers perceived a reduction in the appearance of their pores with the new generation of Laricyl BC10206. This result is higher than for the previous version (60%), even if the difference between the two products is not statistically significant.

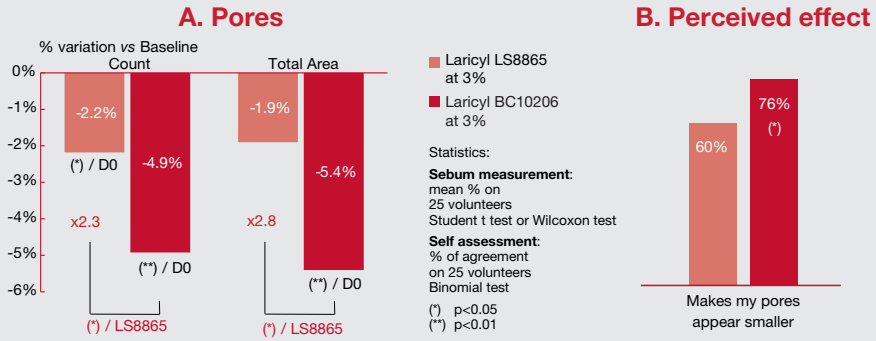
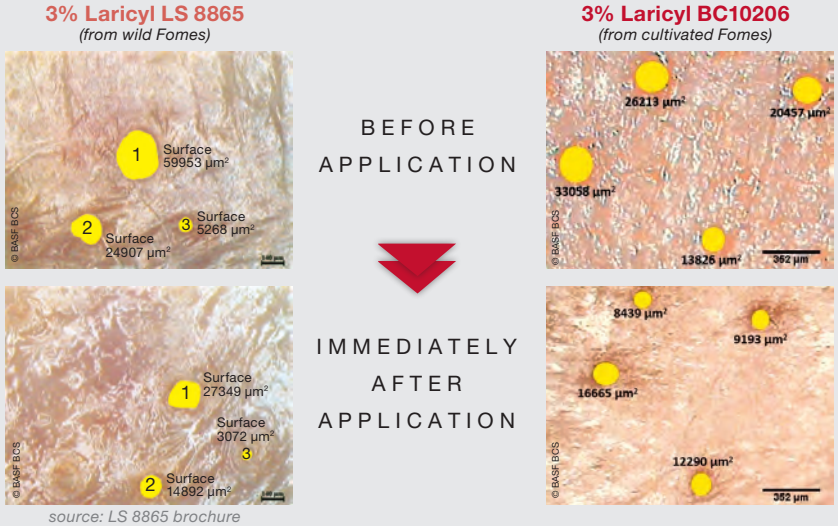


Figure 2 - Pore tightening: immediate effect on pores after application of the new generation of Laricyl BC10206 (from solid state fermentation process), at 3% as compared to the previous Laricyl LS8865 (from mushroom harvested from wild) at 3%. A. Percentage of change of pore count and total pore area. B. Self-assessment.

The reduction of pore size is illustrated by macrophotographs (Fig. 3).



source: LS 8865 brochure

Figure 3 - Illustrative macro-images before and after application of Laricyl.

in vivo

CONCLUSION

In vivo, the new generation of Laricyl minimized the appearance of visible pores immediately after application, with an efficacy 2 times better than the previous version of Laricyl.

MATERIALS&METHODS

Study design

The clinical study was carried-out in double-blind and was randomized with standardized single application of the products by hemi-face. The efficacy of Laricyl BC10206 (from solid state fermentation process) at 3% was compared to the previous version of Laricyl LS8865 (from mushroom harvested from wild) at 3%. Both products were dissolved in water. The study was conducted with a single application, with check points before product application (baseline) and immediately after product application. "Immediately after application" corresponds to the time needed to technically make the measurements and file the self-assessment questionnaire. After application of the products, the technician went straight back to measurements with no time waiting. This time to make the measurements is estimated at ~5-10 min. Note: during this clinical study, the effect of Laricyl application on skin oiliness and shininess was also determined (see hereinafter).

Inclusion criteria

This study was performed on 25 female volunteers in New York state (USA), male and female, aged from 23 to 63 years old (mean 47), with visible pores as assessed by a clinical scientist and oily skin.
Two oily-skinned volunteers with large visible pores were used for illustrative macro-images of pores on the nasolabial fold and nose area, both at baseline and immediately after application of Laricyl BC10206 at 3%.

Application modality

Volunteers were provided with a cotton round soaked with 5ml of either Laricyl BC10206 at 3% or Laricyl LS8865 at 3% and were instructed to swipe the pad over the entire right or left side of their face, as randomly assigned.

Evaluation methods

VISIA Imaging and Image analysis for pore count and total area

The VISIA has a multi-point positioning system and ghost imaging capability that captures well registered images to document progress over time. The VISIA captures and automatically analyzes left, right, and frontal facial views. Visia images were analyzed with Canfield Vaestro software for pore count and pore total area.

Illustrative macro-images of pores

The Trichoscience scope is an image system consisting of a camera with high levels of magnification which allows it to capture detailed images of features of the skin. The Trichoscience Scope with a 60x magnification was used to take illustrative macro-images of pores, both at baseline and after using Laricyl BC10206 at 3%. Pore images were taken in the nasolabial fold and nose region with large pores. ImageJ/Fiji was used to quantify pore area on images captured before and after treatment.

Consumer Self-Perception Questionnaire

The volunteers completed self-administered questionnaires to evaluate their perception of the appearance of their facial pores. Questionnaires were completed through Internet-based software immediately after application. For each Item, the possible answers are: 1: Disagree strongly, 2: Disagree somewhat, 3: Disagree slightly, 4: Agree slightly, 5: Agree somewhat, 6: Agree strongly.

Statistics

Instrumental measurement

The results are then expressed as the mean percentage of change versus baseline (before treatment): a negative value means a reduction of the pore appearance. The statistical analyses of the evolution of the parameters as a function of time and for the product comparison have been done after the verification of the normality of distribution using Shapiro-Wilk test. The Student t test was used if the normality of distribution had been confirmed, in the opposite case, the Wilcoxon test was used.

Volunteer self-perception

To evaluate the efficacy and the appreciation of the products, two percentages Z1 and Z2 are calculated as follows: Z1 = favorable opinion (agree) and Z2 = unfavorable opinion (disagree). The statistical difference in frequencies (%) between favorable (agreement) and unfavorable (disagreement) opinions is evaluated using Two-tailed binomial test at 5%.

EFFICACY

Astringent sensation

in vivo

OBJECTIVE

There is a consumer expectation of a sensation upon the skin resulting from the use of an astringent product. As astringents are substances which causes a tightening or contracting of soft tissue, they can provide a pulled, puckering, or tight feeling upon the skin.

The clinical study was conducted to evaluate the perceivable immediate astringent sensation provided by Laricyl BC10206 at 3% as compared with a placebo.
The study was a double blind, placebo-controlled, split-face and randomized comparative study on 31 female volunteers. The astringent sensation of both Laricyl and the placebo was evaluated immediately after application using a consumer self-perception questionnaire.

RESULTS & DISCUSSION

Laricyl BC10206 was rated as providing a significantly greater ($p<0.05$) astringent sensation on the skin immediately after application as compared to the placebo (Fig. 4).

The average rating by the volunteers for Laricyl was 1.5, it corresponds to a moderate perceived sensation without a too strong uncomfortable feeling.

CONCLUSION

In vivo, the new generation of Laricyl was able to produce a perceptible astringent sensation on the skin.

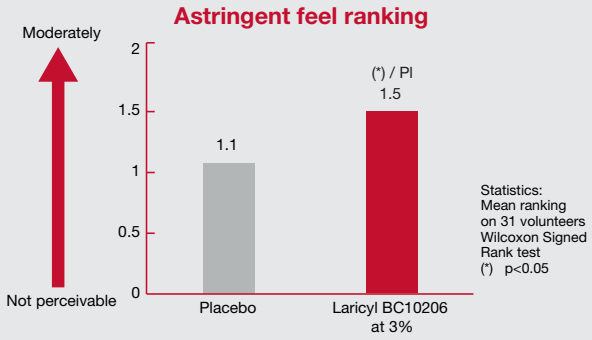


Figure 4 - Astringent sensation perceived immediately after application of the new generation of Laricyl, BC10206 (from solid state fermentation process), at 3% as compared to placebo. Ranking by volunteers' self-evaluation on a 4-point scale.

MATERIALS&METHODS

Study design

The clinical study was carried-out in double-blind and was randomized with standardized single application of the products by hemi-face. The efficacy of Laricyl BC10206 (from solid state fermentation process) at 3% in water was compared to the placebo (water). The study was conducted with a single application, and the astringent sensation was evaluated using a consumer self-perception questionnaire immediately after product application.

Inclusion criteria

This study was performed on 31 female volunteers in New York state (USA), male and female, aged from 23 to 65 years old (mean 49) able to perceive astringent sensation on the skin of the face.

Application modality

Volunteers were provided with a round cotton soaked with 5ml of either Laricyl BC10206 at 3% in water or placebo (water) and were instructed to swipe the pad over the entire right or left side of their face, as randomly assigned.

Evaluation method

Astringent sensation rating by consumer self-perception questionnaire

The perception of astringent sensation was rated on a 4-point scale with the possible answers: (0) Not perceptible, (1) Barely perceptible, (2) Moderately perceptible, (3) Dramatically perceptible.

Statistics

The analysis involves calculating the average rating amongst all panelists for each product. Statistical differences between the ratings for Laricyl BC10206 and the placebo are evaluated using the Wilcoxon Signed Rank Test.

EFFICACY

Removal of excess oil for less shiny skin

in vivo

OBJECTIVE

The clinical study was run to measure the effect on skin oiliness and shininess after the application of new generation of Laricyl BC10206 (from solid state fermentation process), at 3% as compared to the previous Laricyl LS8865 (from mushroom harvested from wild), immediately after application on the face.
The study was a double blind, split-face and randomized comparative study on 25 female volunteers having visible pores and oily skin. The effect on skin oiliness and shininess of Laricyl BC10206 was evaluated by measurement of sebum, illustrative pictures (VISIA) and self-perception questionnaires.

RESULTS & DISCUSSION

Immediately after application of the new Laricyl BC10206, a significant decrease ($p<0.001$) of 43% of the sebum level was measured over the baseline, similar to the decrease observed with the previous Laricyl generation (39%, $p<0.001$) (Fig. 5A).

The visible reduction of skin shine resulting from this less oily facial skin is illustrated by the pictures of 4 volunteers (Fig. 6).

These results were confirmed by the volunteer's self-assessment (Fig. 5B). After both Laricyl BC10206 and LS8865 application, a significant majority ($p<0.05$) of the volunteers perceived a reduction of the excessive shine of their skin (respectively 88% and 72%) and that they leave their skin looking and feel less oily (respectively 84% and 84%).

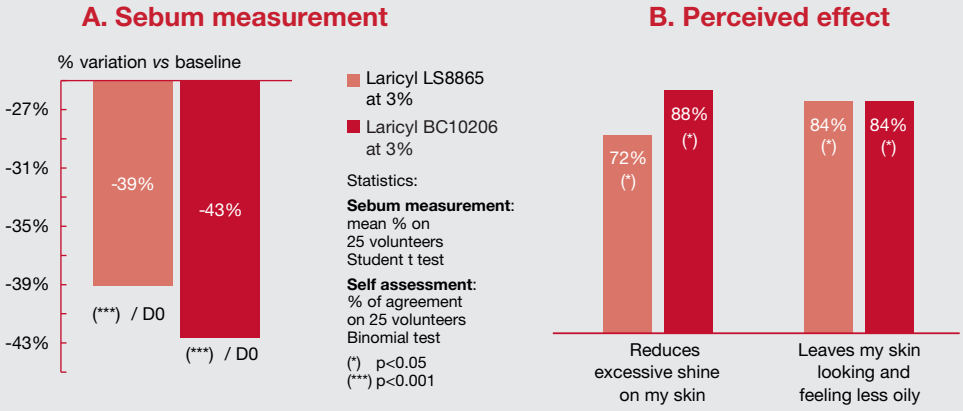


Figure 5 - Sebum level: immediate effect after application of the new generation of Laricyl, BC10206 (from solid state fermentation process) at 3% as compared to the previous Laricyl LS8865 (from mushroom harvested from wild) at 3%. A. Percentage of change of the sebum level vs baseline. B. Self-assessment.

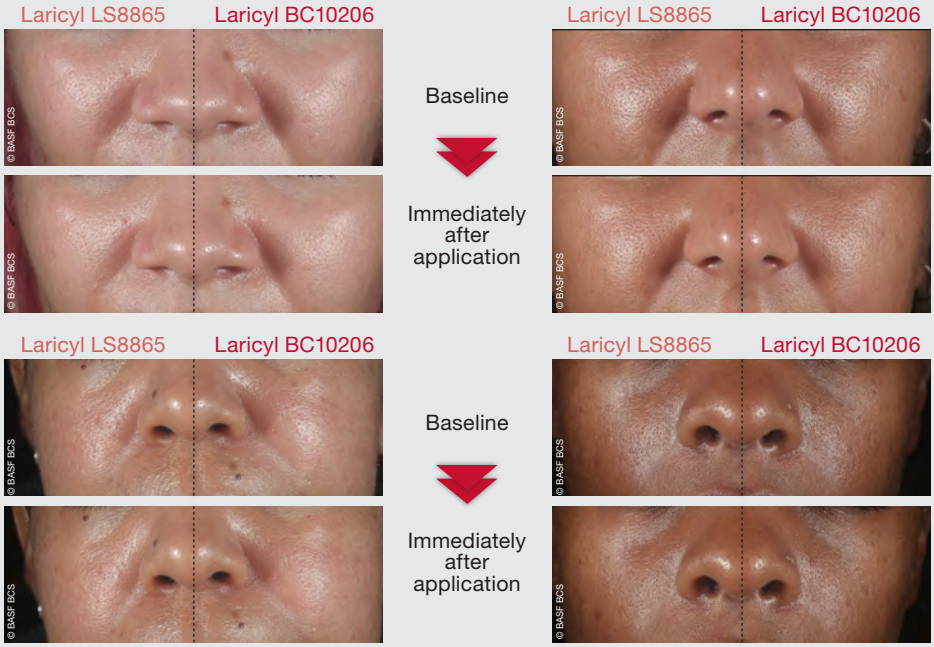


Figure 6 - Illustrative pictures before and after application of Laricyl (previous and new generation).

ex vivo

CONCLUSION

In vivo, the new generation of Laricyl helped to lower excess oil for a less shiny skin.

MATERIALS&METHODS

Study design

The clinical study was carried-out in double-blind, and was randomized with standardized single application of the products by hemi-face. The efficacy of Laricyl BC10206 (from solid state fermentation process) at 3% was compared to the previous Laricyl LS8865 (from mushroom harvested from wild) at 3%. Both products were dissolved in water. The study was conducted with a single application, with check points before product application (baseline) and immediately after product application. "Immediately after application" corresponds to the time needed to technically make the measurements and file the self-assessment questionnaire. After application of the products, the technician went straight back to measurements with no time waiting. This time to make the measurements is estimated at ~5-10 mins.

Note: during this clinical study, the effect of Laricyl application on pore tightening was also determined (see hereinabove).

Inclusion criteria

This study was performed on 25 female volunteers in New York state (USA), male and female, aged from 23 to 63 years old (mean 47), with visible pores, as assessed by a clinical scientist, and oily skin.

Application modality

Volunteers were provided with a cotton round soaked with 5ml of either Laricyl BC10206 at 3% or Laricyl LS8865 at 3% and were instructed to swipe the pad over the entire right or left side of their face, as randomly assigned.

Evaluation methods

Sebum measurement by Sebumeter

The sebumeter SM 815 is designed to measure sebum levels on the skin. A special tape stretched over the measuring head was pressed to the measurement area on the forehead – it was held for 30 seconds to ensure that sebum was absorbed into the tape. Then, the measurement head was inserted into the measurement base, where the transparency of the tape was measured. The more transparent the tape, the higher the sebum level measured.

Illustrative VISIA images of the face

The VISIA has a multi-point positioning system and ghost imaging capability that captures well registered images to document progress over time. The VISIA captures and automatically analyzes left, right, and frontal facial views.

Consumer Self-Perception Questionnaire

The volunteers completed self-administered questionnaires to evaluate their perception of the test products' performance and the resulting skin benefits. Questionnaires were completed through Internet-based software immediately after application. For each Item, the possible answers are: 1: Disagree strongly, 2: Disagree somewhat, 3: Disagree slightly, 4: Agree slightly, 5: Agree somewhat, 6: Agree strongly.

Statistics

Instrumental measurement

The results are then expressed as the mean percentage of change versus baseline (before treatment): a negative value means a reduction of the sebum level. The statistical analyses of the evolution of the parameter as a function of time and for the product comparison have been done after the verification of the normality of distribution using Shapiro-Wilk test. The Student t test was used if the normality of distribution had been confirmed, in the opposite case, the Wilcoxon test was used.

Volunteers self-perception

To evaluate the efficacy and the appreciation of the products, two percentages Z1 and Z2 are calculated as follows: Z1 = favorable opinion (agree) and Z2 = unfavorable opinion (disagree). The statistical difference in frequencies (%) between favorable (agreement) and unfavorable (disagreement) opinions is evaluated using Two-tailed binomial test at 5%.

EFFICACY

Maintenance of skin hydration

ex vivo

OBJECTIVE

As astringents can sometimes be drying, an *ex vivo* study on human *stratum corneum* was conducted to evaluate the effect of Laricyl on skin hydration. Dielectric conductivity of human *stratum corneum* treated with the new generation of Laricyl BC10206 (from solid state fermentation process) or the previous Laricyl LS8865 (from mushroom harvested from wild) in hydrogels was compared to the placebo hydrogel.

RESULTS & DISCUSSION

Laricyl BC10206 and Laricyl LS8865, both applied at 3% and 5% in hydrogel, increased significantly the dielectric conductivity of the human *stratum corneum* versus the placebo hydrogel (Fig. 7). The hydration effect over 24h was of +40% (SEM ± 6%, $p<0.01$) and +100% (SEM ± 22%, $p<0.001$) for the new generation Laricyl BC10206 at respectively 3% and 5% doses. These results are similar to the one obtained with the previous Laricyl LS8865 (+47% ± 16%, $p<0.05$, and +102% ± 14%, $p<0.001$, respectively at 3% and 5% dose).

These results indicate that Laricyl BC10206 will contribute to preserve the skin moisturization without drying the skin.

Conductimetry on *stratum corneum* (hydration)

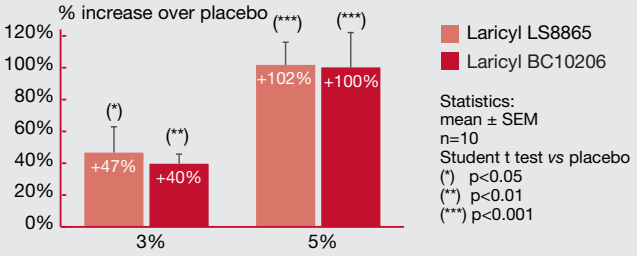


Figure 7 - Hydration of the human *stratum corneum* by conductimetry after application of the Laricyl BC10206 and LS8865 at 3% and 5%, and placebo. Conductimetry measurements at 0.5, 2, 2, 4, 6 and 24h after application, calculation of the area under the curve and determination of the % of change over the placebo control.

CONCLUSION

Ex vivo, the new generation of Laricyl increased the dielectric conductimetry of human *stratum corneum*, and thus may contribute to maintain the skin moisturization.

MATERIALS&METHODS

***Stratum corneum* preparation**

This study was conducted with human skin explants from plastic surgery. Epidermal sheets were prepared by peeling the dermis after heating at 60°C for 2 minutes. The *stratum corneum* was isolated by enzymatic digestion as described [Rochefort *et al*, 1986].

Treatment

Epidermal sheets were balanced in an oven set at a relative humidity (44% - saturated solution of potassium carbonate), then were treated with the hydrogels (composition: Carbomer 0.60%, Chlorphenesin 0.30%, Water qsp 100%; pH 5.8) containing Laricyl BC10206 at 3% or 5%, Laricyl LS885 at 3% or 5%, or the placebo hydrogel as control. 1 mg/cm² of each hydrogel was applied three times at 30 min intervals before measurements.

Dielectric conductivity measurement - Tagami test [Obata & Tagami, 1989; Obata & Tagami, 1990]

Water content on *stratum corneum* were measured by the dielectrical conductivity (micro-Siemens, µS) method using a hygrometer Skicon 200 (IBS Inc., Hamamatsu, Japan). The principle of the method is based on the relationship between variation of skin dielectrical conductivity and its water content, due to the bipolarity of water molecules. The more the *stratum corneum* is hydrated, the higher conductivity is. Dielectric conductivity measurements were carried out prior to treatment and then at 0.5, 1, 2, 4, 6 and 24 hours after treatment. Measurements were performed in triplicates on each sample and for each time interval.

Results and statistics

The experiments were performed on 10 *stratum corneum* samples from 3 donors. The moisturizing effect was calculated by determining the area under the dielectric conductivity curve over 24 hours. The results are expressed as the percentage ± SEM of the average area under the curve for the tested product *vs* the placebo control. The statistical evaluations were calculated based on Student t test after validation of the normality and the equality of the variances.



GENERAL CONCLUSION

New generation of Laricyl that features a more sustainable manufacturing process while offering proven enhanced pore minimizing efficacy.

- Upgraded version from *Fomes officinalis* mushroom mycelium produced by solid-state fermentation.
- Clinically proven immediate pore-tightening effect while contributing to reduce skin shine and to maintain skin moisture.

People commonly complain about their large visible skin pores on their face which affect their beauty. They find them too dilated and mostly associated with oily and over-shinning skin.

For 20 years, Laricyl, an extract from the *Fomes officinalis* mushroom, has been a tried-and-tested component of BASF's portfolio of cosmetic active ingredients for pore refinement. For more sustainable practices, the sourcing from wild of the mushroom has been replaced by a biotechnological process of solid state fermentation to produce the mushroom mycelium, that is now organic certified. This sustainable revision of the ingredient is in line with the goals of the company's Responsibly Active program for botanical supply chains.

In a double-blind, split-face, comparative and randomized study with 25 female subjects, the new generation of Laricyl at a concentration of 3% improved the appearance of pores in size and number. Image analysis showed that it minimized the appearance of visible pores immediately after application 2.8 times better than the previous version of the ingredient. A perceived astringent sensation has also been assessed in a double-blind, placebo-controlled, split-face and randomized study on 31 volunteers. The active ingredient also helped to remove excess oil from the skin, making it less shiny without drying it out.

ANNEX

Technical data - Available upon request

- Quality and regulatory product information
- Product purity profile
- Toxicological abstract
- Composition sheet
- Specifications
- Technical information
- Vegan statement
- Natural content origin (according to ISO 16128)

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Europe

BASF Beauty Creations

49, avenue Georges Pompidou
92593 Levallois-Perret Cedex
France

Phone: +33 (0) 1 49 64 52 09
bcs-europe@basf.com

North America

BASF Corporation
North American Regional Headquarters
100 Park Avenue
Florham Park, NJ 07932
USA

Phone: +1-800-962-7831
CosmeticsCustomerCare@basf.com

South America

BASF S.A.
Av. das Nações Unidas, 14171
Crystal Tower
04794-000 - São Paulo - SP
Brazil

Phone: +55 11 2039 2273
personal-care-sa@basf.com

Asia Pacific

BASF East Asia
Personal Care
45/F Jardine House
No.1 Connaught Place
Central, Hong Kong

Phone: +852 2731 0190
personal-care-hk@basf.com

Publisher
BASF Beauty Care Solutions France SAS
3, Rue de Seichamps - 54425 Pulnoy (France)
Email: bcs-europe@basf.com
www.personalcare.basf.com

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